

RevPlay – ER Diagram (Database Relationships)

This document describes the Entity Relationship Diagram (ERD) for the RevPlay console-based music streaming application. It outlines entities, attributes, and relationships used to ensure data integrity and scalability.

Entities and Relationships

Entity	Key Fields	Relationships
Users	user_id (PK), email, password_hash, role, security_question, security_answer_hash	1-1 Artists, 1-M Songs, 1-M Albums, 1-M Playlists
Artists	artist_id (PK), user_id (FK), bio, genre, social_links	1-1 Users
Songs	song_id (PK), title, genre, duration, play_count, album_id (FK)	M-1 Users (Artist), M-1 Albums
Albums	album_id (PK), title, release_date, artist_id (FK)	M-1 Users (Artist), 1-M Songs
Playlists	playlist_id (PK), name, description, is_public, user_id (FK)	M-1 Users, M-M Songs
Playlist_Songs	playlist_id (FK), song_id (FK)	M-M Playlists and Songs
User_Favorites	user_id (FK), song_id (FK)	M-M Users and Songs
Listening_History	history_id (PK), user_id (FK), song_id (FK), played_at	M-1 Users, M-1 Songs

Design Notes

- Users act as the central authentication entity.
- Artists extend users via a one-to-one relationship.
- Albums and songs are owned by artists.
- Many-to-many relationships are handled using junction tables.
- Foreign keys enforce ownership and data integrity.